# Run your own Universal POS server (0.1.9)

> **Internal only.** Self-host is not offered to clients (owner, 2026-10-20). Use this package for your own servers (for example a cloud VPS); never give the `.deb` to a client.

For the person who looks after the shop's computer. One package installs the sync server, the shop web panel and HTTPS. Your tills keep selling when this server is off; they catch up when it is back.

## 1. What you need

| Item | Requirement |
|---|---|
| Computer | Ubuntu 22.04 or 24.04, or Debian 12; 64-bit Intel/AMD; 2 CPU cores, 4 GB RAM, 20 GB free disk |
| Internet | Needed during install (to fetch PostgreSQL). The tills need to reach the server: same Wi-Fi, a domain, or Tailscale |
| Seller contact | WhatsApp or call +91 89229 62849, universalpos2026@gmail.com |
| From your seller | `universal-pos-server_<version>_amd64.deb`, your licence file (`.poslicense`), optionally a phone-home token |
| Ports | 80 and 443 free (domain and LAN modes) |

## 2. Install

```
 sudo apt update
sudo apt install ./universal-pos-server_0.1.9_amd64.deb
```

This installs PostgreSQL too. It ends with "Next: sudo pos-server setup".

## 3. Set up

```
sudo pos-server setup
```

It asks:

| Question | Answer |
|---|---|
| Access mode | `domain` (a web address you own, e.g. `pos.example.com`, pointing at this computer; free HTTPS certificate), `lan` (only on the shop Wi-Fi), or `tailscale` (reach it from anywhere through Tailscale, no domain) |
| Domain / LAN address | For `domain` the address; for `lan` the computer's Wi-Fi address (it suggests one) |
| Licence file | Path to the `.poslicense` from your seller. Your shop is created with the same ids as on your phone, so the phone can join without losing data |
| Shop admin e-mail and password | The web panel login (8+ characters) |
| Phone-home token | Optional, from your seller: renewals then arrive by themselves |

At the end it prints the server address, the web panel address and a pairing code (15 minutes, single use).

**Tailscale mode** needs Tailscale installed and logged in first: `curl -fsSL https://tailscale.com/install.sh | sh`, then `sudo tailscale up`.

**LAN mode** uses the server's own certificate. Each phone and PC trusts it once: copy `/var/lib/universal-pos/caddy/caddy/pki/authorities/local/root.crt` to the device and install it as a trusted certificate (Android: Settings → Security → Encryption & credentials → Install a certificate → CA certificate).

Run `sudo pos-server setup` again any time to change the access mode; your data is never touched.

Without questions (scripts):

```
sudo pos-server setup --non-interactive --mode domain --domain pos.example.com \
  --licence shop.poslicense --admin-email owner@example.com --admin-password-file
pw.txt
```

## 4. Connect the tills and the web panel

- Phone already set up as the shop: **Settings** → **Cloud sync** → **Server address** = the address setup printed → **Pairing code** → **Pair**.
- New phone: **Join existing shop** → address + code + its own invoice series (e.g. `A2`) → **Join**.
- A new code for each device: `sudo pos-server pair-code`.
- Web panel: open the address in a browser, sign in with the shop admin e-mail and password.

## 5. Everyday commands

| Task | Command |
|---|---|
| Is everything running? | `sudo pos-server status` |
| Pairing code | `sudo pos-server pair-code` (several outlets: `--shop <id>`) |
| Renewed licence file | `sudo pos-server import-license new.poslicense` |
| Another outlet | `sudo pos-server add-shop outlet2.poslicense`, then `pair-code --shop <id>` |
| Another web panel login | `sudo pos-server create-admin --email manager@example.com` |
| Logs | `sudo pos-server logs -f` |
| Renewals now (with a token) | `sudo pos-server vendor-checkin` |

## 6. Updates

```
sudo apt install ./universal-pos-server_<new version>_amd64.deb
```

The update stops the server, backs up the database, updates it, and starts the new version. If anything fails it puts the backup and the old version back by itself and says why; your tills keep working throughout. The last 3 versions stay installed.

Back to the previous version by hand: `sudo pos-server rollback` (asks first; restores the backup taken before the update, so sales made on this server since then would be lost — tills still have theirs and send them again).

## 7. Backups

- Automatic every night at 02:30, the last 7 kept, in `/var/backups/universal-pos/`. Copy that folder to a USB drive or another computer now and then.
- Now: `sudo pos-server backup`.
- Restore: `sudo pos-server restore /var/backups/universal-pos/<file>.dump` (asks first; saves the current database before replacing it).

## 8. Uninstall

| Command | Removes | Keeps |
|---|---|---|
| `sudo apt remove universal-pos-server` | Programs and services | Database, settings, backups |
| `sudo apt purge universal-pos-server` | Also settings and logs | Database and backups (it prints how to delete them if you really want to) |

## 9. Problems

| Symptom | Fix |
|---|---|
| Setup: "port(s) 80, 443 are in use" | Another web server runs on this computer: stop it ( `sudo systemctl stop apache2` or `nginx` ), or use `tailscale` mode |
| Setup: "PostgreSQL is not running" | `sudo systemctl start postgresql` |
| Domain mode: browser says the certificate is not valid | The domain must point at this computer's public address and ports 80/443 must reach it (router port forwarding) |
| Tills say "Server not reachable" | `sudo pos-server status` ; check the address on the till; in LAN mode the till must be on the same Wi-Fi |
| Pairing: "This shop has no license yet" | `sudo pos-server import-license <file>` |
| An update failed | It rolled back by itself; send `/var/log/universal-pos/upgrade.log` and `sudo pos-server logs` to your seller |